- A mature schizont contains 8–32 merozoites and haemozoin. The mature schizont bursts releasing the merozoites into the circulation.
- The merozoites invade fresh erythrocytes within which they go through the same process of development. This cycle is called erythrocytic schizogony.
- The rupture of the mature schizont releases large quantities of pyrogens. This is responsible for the febrile paroxysms characterising malaria.
- In *P. falciparum*, erythrocytic schizogony always takes place inside the capillaries and vascular regions of internal organs. Therefore, in these infections, schizonts and merozoites are usually not seen in the peripheral blood.

C. Gametogony

- Some of the merozoites, after a few erythrocytic cycles do not develop into trophozoites and schizonts but they undergo sexual differentiation to develop into the gametocytes.
- Development of gametocytes takes place within the internal organs and only the mature forms appear in circulation.
- The mature gametocytes in *P. falciparum* are crescent shaped.
- Female gametocytes are generally more numerous and larger.
- Male gametocytes and female gametocytes are called micro gametocytes and macro gametocytes respectively.
- Gametocyte appears in 10–12 days in *P. falciparum*.
- The gametocytes do not cause any clinical illness in the host, but are essential for transmission of the infection.
- A person who harbors the gametocytes is referred to as a carrier or reservoir.

Infobits

Sir Ronald Ross Institute of Parasitology is a malaria research institute located in Begumpet, Secunderabad, Hyderabad, India. Established in 1955, the institute is a division of Osmania University. The institute is named after Sir Ronald Ross, winner of Nobel Prize for Physiology or Medicine, 1902. Though he was a surgeon by qualification, Ross was attracted towards research in tropical diseases, especially malaria. During his posting, he worked on his research from a laboratory in the old Begumpet military hospital building. It was in this building on 20 August 1897 that he made the discovery of the malarial parasite inside the body of a mosquito.His study confirmed that mosquitoes were the carriers of malaria parasite.

8.5.6 Mosquito Cycle (Sexual Cycle – Sporogony)

- A Female Anopheles mosquito during its blood – meal from an infected person, sucks up both the sexual and asexual forms of parasite. But, only the mature sexual forms develop and the rest die.
- The gametocytes are set free in the midgut (stomach) of mosquito and undergo further development.

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